A. Sizing code verification: % errors

1. Design point output aircraft verification % errors (using book takeoff gross weight)

For the table below, the takeoff gross weight stays fixed at the true value for each aircraft. The empty weight of the SR-71 was tweaked by tweaking the active cooling weight which is an unknown because it is not published.

- TOGW = Takeoff gross weight
- W_e = Empty weight
- W_f = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

Table 1: Percent errors of predicted performance parameters relative to reference aircraft data (using true takeoff gross weight).

Aircraft	*TOGW*	W_e	W_f	*Alt*	*R*	*L/D*	*TSFC*
			Supersonic fighters				
XB-70	Book	1.5	0.6	Book	Book	5.3	4.5
SR-71	Book	0.6	5.1	Book	Book	2.3	2.8
			Supersonic transports				
Concorde	Book	0.1	5.3	Book	Book	0.3	0.8
TU-144D	Book	4.9	4.6	Book	Book	2.5	2.7
			Subsonic transports				
B737-200	Book	4.7	3.5	Book	Book	1.4	2.6
B777-200er	Book	4.1	4.3	Book	Book	0.8	2.3
A320-200	Book	1.8	4.4	Book	Book	1.5	3.9
2000 EX	Book	6.8	1.3	Book	Book	0.6	0.3
			Subsonic general aviation				
CJ3	Book	0.3	2.0	Book	Book	0.7	6.8

2. Geometric wing output verification % errors (using book takeoff gross weight)

The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- b = wingspan
- c_r = root chord length
- Λ = leading edge sweep angle

Table 2: Percent errors of predicted wing geometric variables relative to reference aircraft data (using true takeoff gross weight).

Aircraft	*S*	*b*	c_r	Λ
		Supersonic fighters		
XB-70	2.2	4.3	1.9	—
SR-71	3.4	1.2	5.1	—
		Supersonic transports		
Concorde	0.7	0.2	12.3	—
TU-144	0.3	0.0	1.7	—
		Subsonic transports		
B737-200	1.2	0.6	4.5	0.0
B777-200er	0.8	0.4	0.4	0.1
A320-200	4.9	2.4	2.4	0.0
2000 EX	0.4	0.2	0.2	—
		Subsonic general aviation		
CJ-3	0.5	0.3	0.3	—

3. Design point output aircraft verification % errors (takeoff gross weight free to iterate)

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect.

- TOGW = Takeoff gross weight
- W_e = Empty weight
- W_f = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

Table 3: Percent errors of predicted performance parameters relative to reference aircraft data (allowing takeoff gross weight to iterate).

Aircraft	*TOGW*	W_e	W_f	*Alt*	*R*	*L/D*	*TSFC*
			Supersonic fighters				
XB-70	6.0	4.2	7.6	4.8	1.0	7.0	2.6
SR-71	5.0	10.7	1.3	8.4	1.2	0.0	3.0
			Supersonic transports				
Concorde	5.1	2.9	7.8	1.1	3.2	2.8	1.4
TU-144D	16.1	13.3	12.4	4.8	1.7	10.7	9.5
			Subsonic transports				
B737-200	8.5	8.5	12.3	8.0	2.0	0.1	2.6
B777-200er	13.2	11.2	11.8	10.8	3.8	7.4	3.9
A320-200	8.7	6.5	14.7	12.6	3.5	7.0	5.9
2000 EX	16.0	12.9	22.2	5.6	4.1	4.5	0.3
			Subsonic general aviation				
CJ3	3.0	2.1	5.1	15.2	2.3	10.6	4.4

4. Geometric wing output verification % errors (takeoff gross weight free to iterate)

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect. The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- b = wingspan
- c_r = root chord length
- Λ = leading edge sweep angle

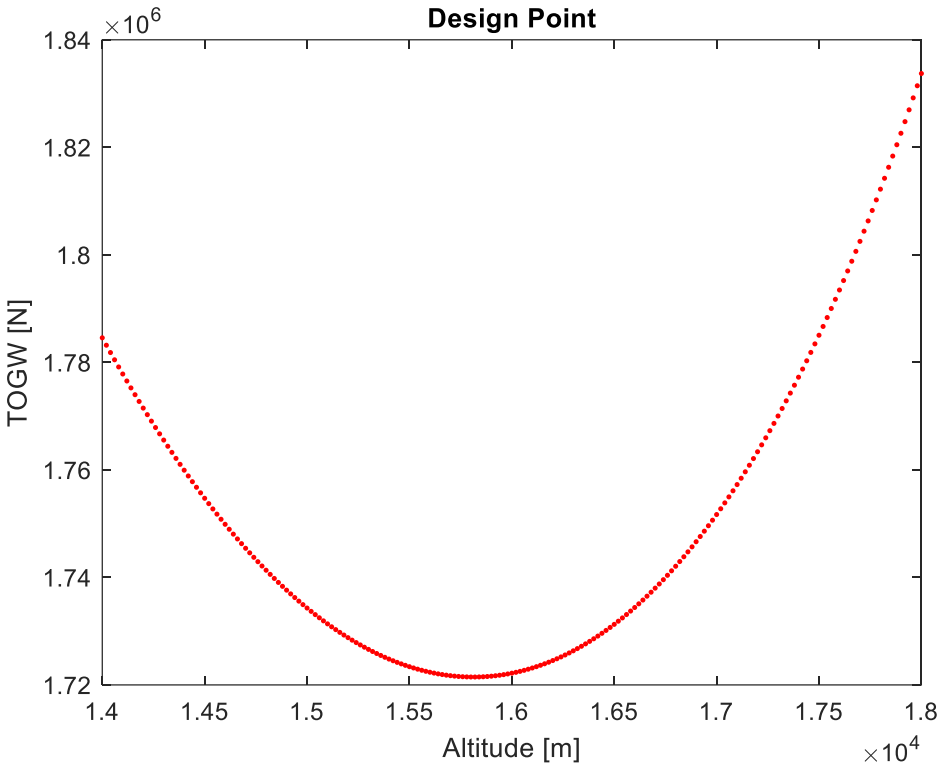
Table 4: Percent errors of predicted wing geometric variables relative to reference aircraft data (allowing takeoff gross weight to iterate).

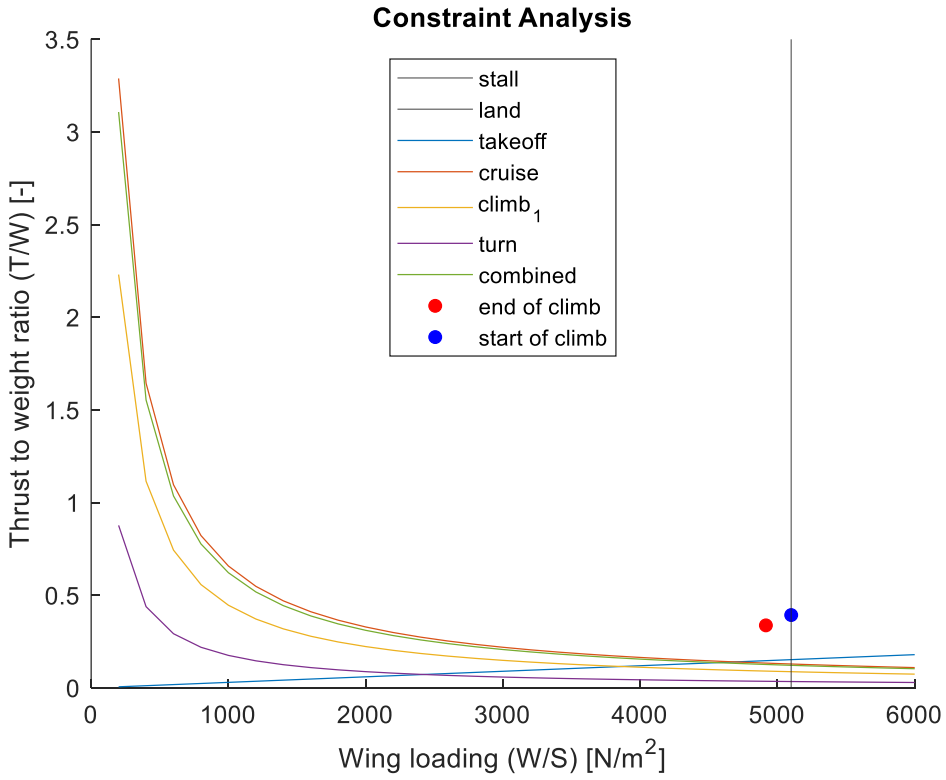
Aircraft	*S*	*b*	c_r	Λ
		Supersonic fighters		
XB-70	8.3	7.4	1.0	—
SR-71	8.5	3.7	7.7	—
		Supersonic transports		
Concorde	5.8	2.8	14.6	—
TU-144	16.4	8.4	10.0	—
		Subsonic transports		
B737-200	7.2	3.5	0.6	0.0
B777-200er	12.5	6.5	6.5	0.1
A320-200	10.2	5.2	5.2	—
2000 EX	16.5	8.0	7.9	—
		Subsonic general aviation		
CJ-3	3.5	1.7	1.8	—

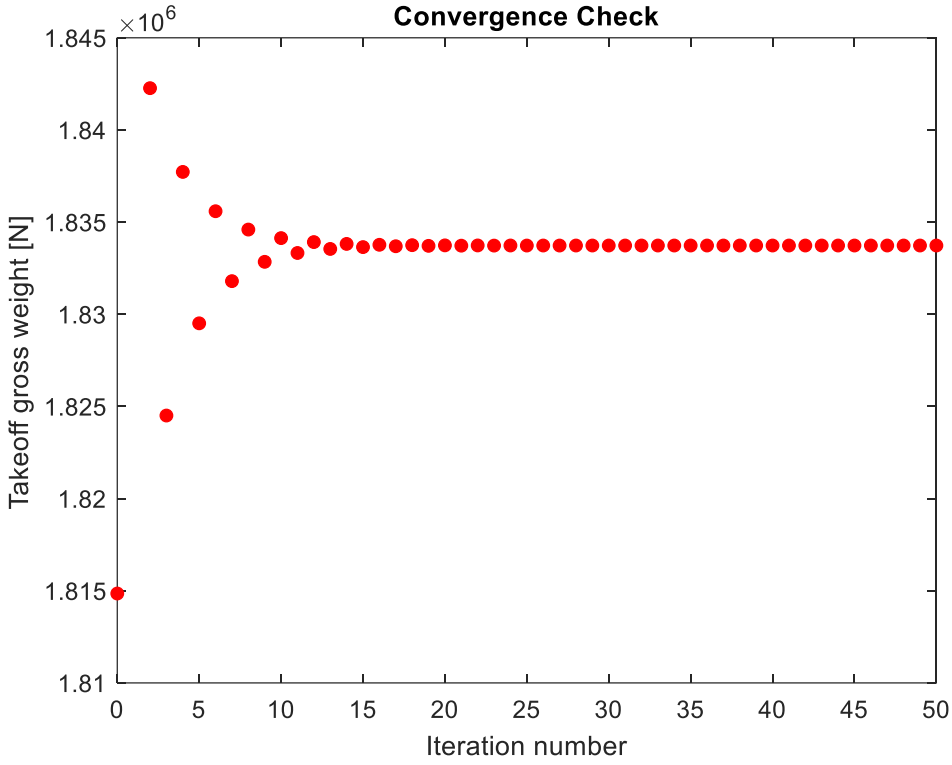
5. Example supersonic aircraft sizing code outputs: Concorde

Model status: These scripts are conceptual/preliminary engineering models. Several aerodynamic, propulsion, structural, and thermal relationships are analytical or empirical approximations. Results should not be interpreted as certification-level aircraft predictions. The purpose of the models is to investigate multidisciplinary aircraft-design relationships.

- TOGW=1721471.554320
- W_e=749858.906496
- Range=6397448.703994
- Lift to drag ratio=6.938186
- Drag_per_engine=46251.797458
- S=337.532853
- Alt=15820.000000
- V=590.136037
- V_a=220.653246
- c_r=23.619170
- c_t=3.542876
- b=24.853272
- Sweep=55.000000
- CPR=15.500000
- BR=0.000000
- FPR=0.000000
- A_inlet=0.951000
- Thrust_required=46251.797458
- TIT=1400.000000
- TSFC=0.000033
- S_error=5.782874
- b_error=2.764977
- c_r_error=14.608929
- TOGW_error=-5.145243
- We_error=-2.873930
- Wf_error=7.811536
- alt_error=1.125000
- R_error=-3.184657
- L/D_error=2.826522
- TSFC_error=1.427438


Design Point
×10^6
1.84
1.82
1.8
1.78
1.76
1.74
1.72
TOGW [N]
1.4
1.45
1.5
1.55
1.6
1.65
1.7
1.75
1.8
Altitude [m]
×10^4


Constraint Analysis
stall
land
takeoff
cruise
climb_1
turn
combined
end of climb
start of climb
3.5
3
2.5
2
1.5
1
0.5
0
Thrust to weight ratio (T/W) [-]
0
1000
2000
3000
4000
5000
6000
Wing loading (W/S) [N/m^2]


Convergence Check
×10^6
1.845
1.84
1.835
1.83
1.825
1.82
1.815
1.81
Takeoff gross weight [N]
0
5
10
15
20
25
30
35
40
45
50
Iteration number

6. Methods used in the sizing code in order:

1. Aerodynamics methodology to obtain the lift to drag ratio

- Obtain subsonic airfoil drag polar data of various airfoils using airfoiltools.com
- Begin the altitude sweep and TOGW convergence for-loops
- Calculate ambient atmospheric properties
- Use a simplified minimum pressure coefficient formula which feeds into the wing sweep solver. The wing sweep solver uses the isentropic pressure-ratio relation to solve for the critical Mach number. This and the Mach number determines the angle of the sweep.
- Solve for average weight during cruise and use it to size the aircraft.
- Calculate wing aspect ratio, wing area, and span, Oswald efficiency factor, root chord length, sectional and overall lift curve slopes. The wing area is calculated from the max takeoff gross weight and the sea level stall speed and air density.
- Calculate engine drag using the frontal area of each engine.
- Calculate the compression lift using the shock properties generated at the nose of the aircraft.
- Solve for the fuselage length (if desired) using the fuel volume as the fuselage volume minus 10 meters of length for non-fuel volume.
- Solve for the coefficient of skin friction using empirical relations
- Solve for the fuselage and wing parasite drag coefficient, zero-lift wave wing drag coefficient, and fuselage wave drag coefficients. Together, these make up the total zero-lift drag coefficient. The zero-lift wave uses a simplifying relation that depends on sweep and total airfoil thickness.
- Solve for the coefficient of lift using the mean weight and compression lift (if activated). For the subsonic case, the airfoil drag polar data is used to obtain the coefficient of lift and drag (adding induced drag in separately later).
- Solve for angle of attacks using the lift-curve slope and coefficient of lift.
- Solve for the coefficient of wave drag due to lift using a simplifying expression which uses the angle of attack (supersonic case).
- Solve for the induced drag coefficient (which I believe goes to 0 for the XB-70 if it applies wing droop).
- Solve for the total drag. For the supersonic case, the total drag is zero lift + induced + wave due to lift + engine drag. For the subsonic case, since drag polar data is used, The total drag is the drag polar drag + induced drag + fuselage parasite drag.
- Calculate the thermal protection system weight (supersonic) that depends on the weight of the insulation type material and the weight of the cooling jacket.
- Finally, calculate the lift to drag ratio that is fed into the Breguet fuel mass formula for the convergence of the TOGW at each altitude.

2. Propulsion methodology to obtain the thrust specific fuel consumption (TSFC)

- An if statement is created that controls which chunk of equations to solve: for the ideal engine or for an existing engine. For the ideal engine, a series of for-loops are created that solves for engine variables like bypass ratio, compressor pressure ratio, and turbine inlet temperature that must produce a thrust equal to the drag of the aircraft per engine.
- The inlet, compressor, fan, and turbine total temperatures and pressures are solved for.
- The fuel flowrate per oxygen flowrate fraction is solved for using the turbine inlet temperature.
- The exit temperature is solved for and is used to calculate the engine core and fan exit velocity.
- A for loop is created to solve for the sea level mass flow rate. The kinetic energy formula (Ke_dot=.5*m_dot*Ve^2) is used to help solve for m_dot.
- Next, the core thrust is solved for and adjusts the turbine inlet temperature to thrust match the drag of the aircraft. If the total turbine inlet temperature becomes maxed out, the afterburner is applied for the remaining thrust.
- The afterburner exit velocity is solved for by using the specified total afterburner temperature which solves for the fuel fraction of the afterburner required to generate the remaining thrust that the core itself cannot generate without the afterburner.
- Next, the thrust specific fuel consumption is solved for.
- Next, the engine weight is solved for (if solving for ideal engine) using Raymer's approximations that depend on the sea level thrust.

3. Structural methodology to obtain weights

- Raymer's approximations are used to determine all the different weight components of the empty weight.
- Next, the empty weight is found by adding the listed weights together plus the skin, thermal protection system (supersonic), and payload weights.

4. Performance and trajectory methodology for climb and descent

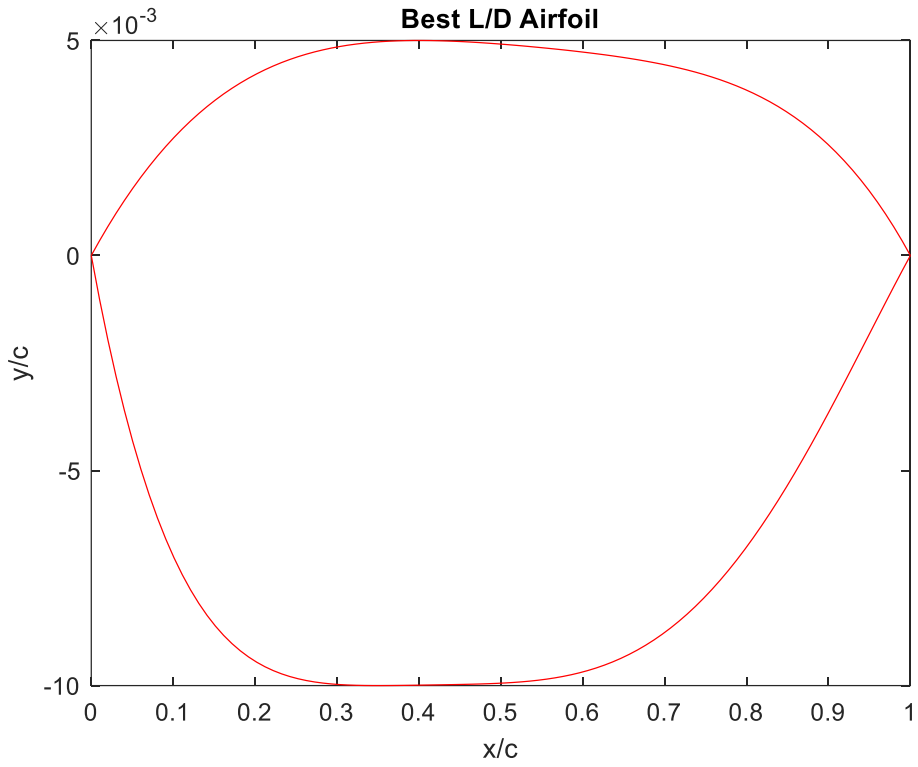
- The total fuel required for the descent weight is found first by starting with the landing weight and sink rate.
- Next, the Breguet fuel mass is calculated using this weight at the end of cruise to determine the weight at the beginning of cruise.
- Next, the total fuel required for the climb phase is calculated by starting with the initial cruise weight and climb rate.
- Finally, the total takeoff gross weight is found by adding the climb weight to the initial cruise weight. This value is the TOGW used for each altitude.

5. Carpet plots (constraint diagrams)

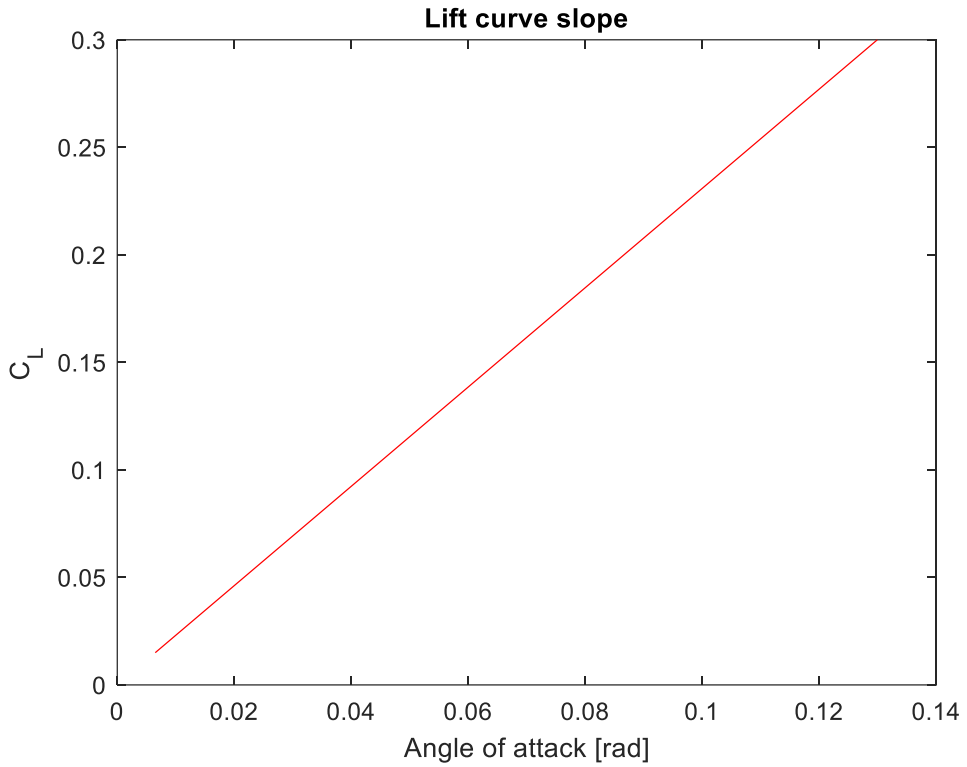
- The wing loading equations for stall and land are specified.
- The thrust to weight ratio for takeoff, climb, cruise, and turn are specified. These equations use the required thrust. The takeoff distance includes clearing a 50ft obstacle.
- Next, the initial climb and initial cruise design point is plotted against these curves. These two points use the available thrust.
- Next, the upper left space defined by the curves is the area for feasibility.

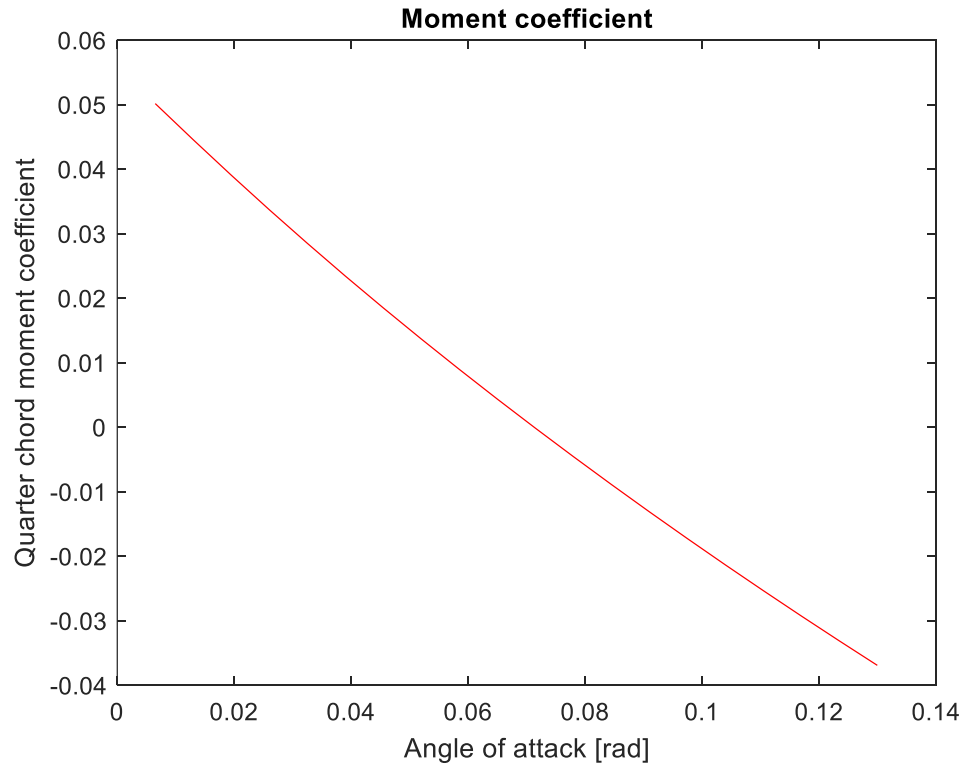
7. Airfoil optimization code outputs for the Concorde flight condition

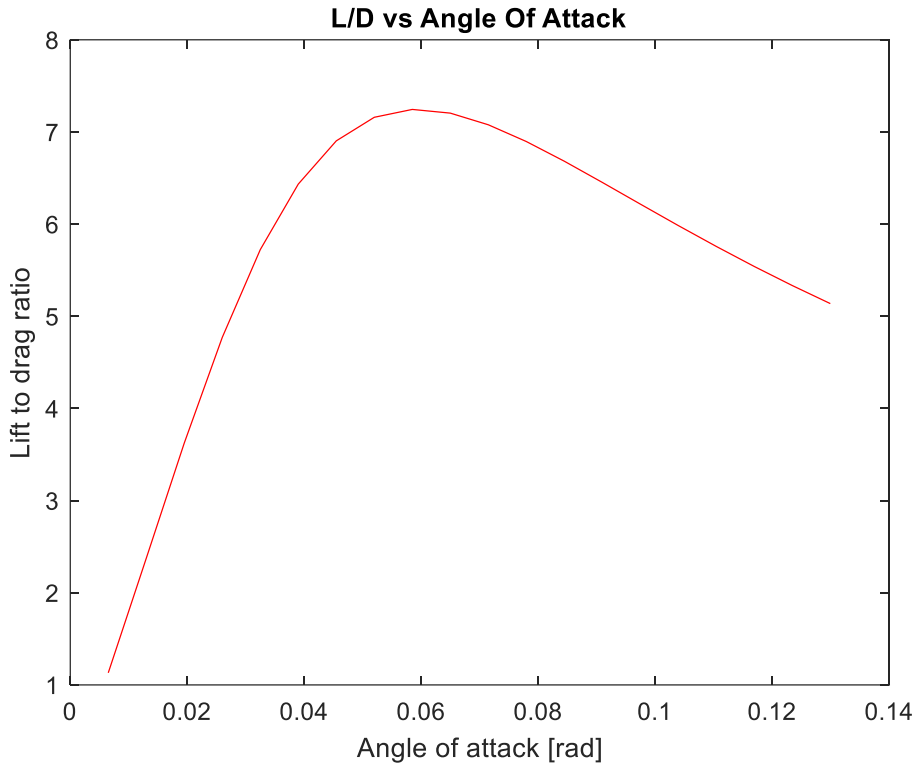
- aoa=0.057201
- x_crest_u=0.400000
- y_crest_u=0.005000
- x_crest_l=0.350000
- y_crest_l=-0.010000
- L/D=7.088380

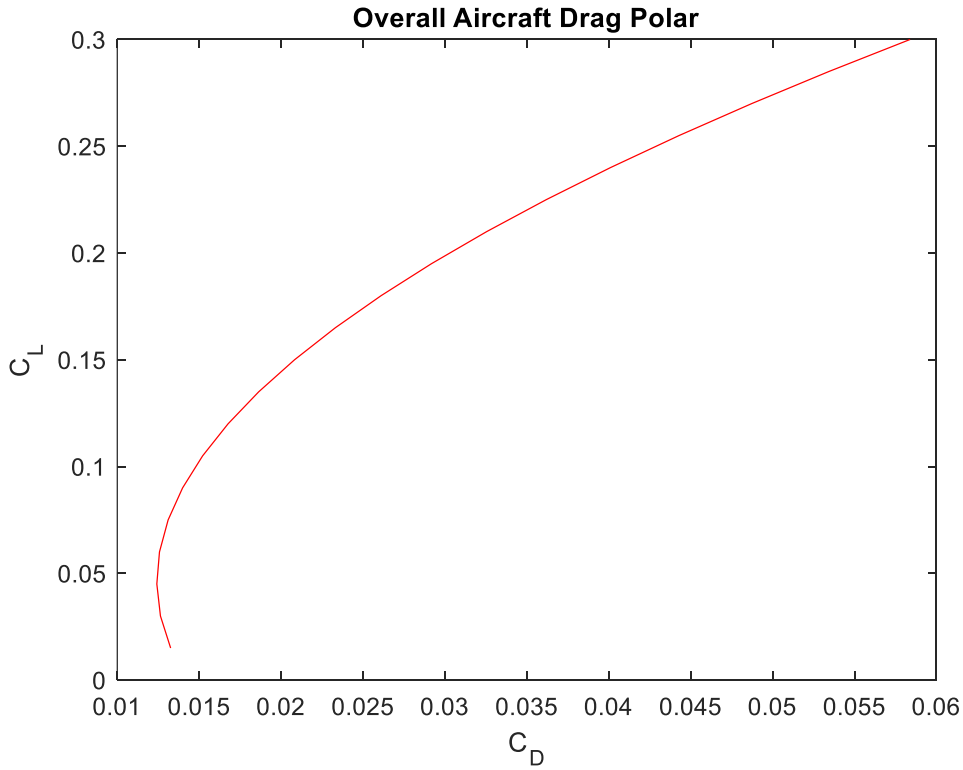


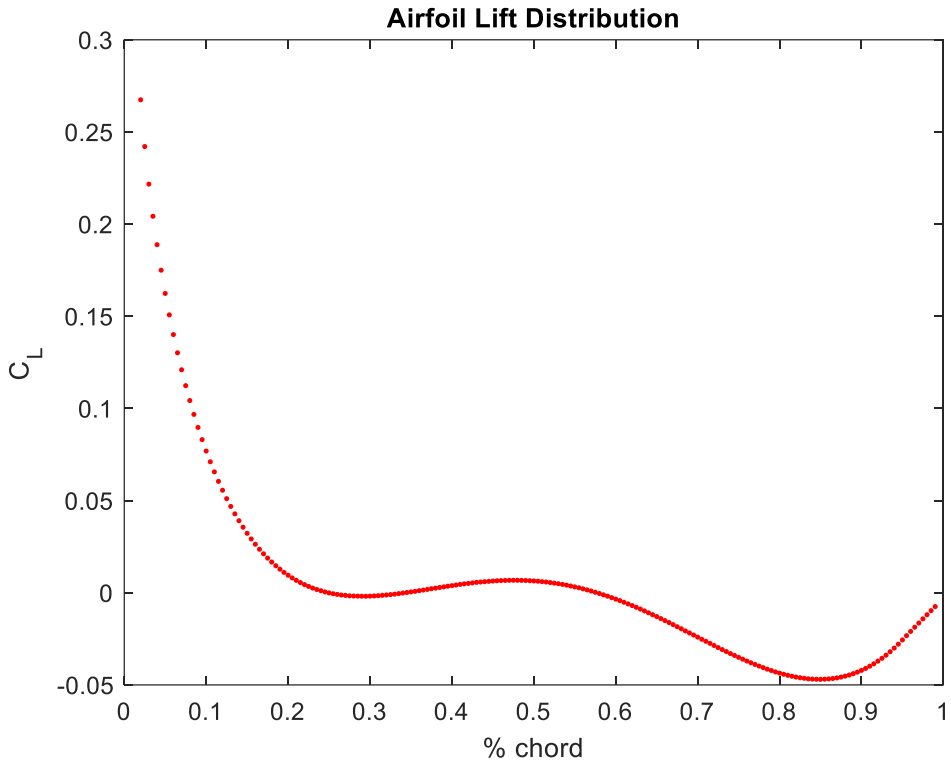
.


Lift curve slope
0.3
0.25
0.2
0.15
0.1
0.05
0
C_L
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
Angle of attack [rad]


Moment coefficient
0.06
0.05
0.04
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
-0.04
Quarter chord moment coefficient
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
Angle of attack [rad]


L/D vs Angle Of Attack
Lift to drag ratio
Angle of attack [rad]
8
7
6
5
4
3
2
1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14


Overall Aircraft Drag Polar
C_L
C_D
0.3
0.25
0.2
0.15
0.1
0.05
0
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
0.05
0.055
0.06


Airfoil Lift Distribution
0.3
0.25
0.2
0.15
0.1
0.05
0
-0.05
C_L
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
% chord

8. Methods used in the airfoil optimization code in order:

- For loops of the max thickness location and max thickness of the upper surface and lower surface air foil are created to iterate for the best combination.
- Next, the coefficients of a polynomial are solved for using the geometric input variables of the airfoil. The subsonic polynomial takes the form y/c = a1*x^(1/2) + a2*x + a3*x^2 + a4*x^3 + a5*x^4. The supersonic polynomial takes the form y/c = a1*x + a2*x^2 + a3*x^3 + a4*x^4 + a5*x^5.
- Next, for the subsonic case, for-loops are created to solve for the A coefficients and zero lift angle of attack of thin airfoil theory. For the supersonic case, shock and expansion relations are used.
- Next, the velocity at each point along the chord is calculated using the A coefficients for the upper and lower surface for the subsonic case.
- Isentropic relations are then used to calculate the pressure coefficient at each point along the chord.
- Next, the axial and normal force coefficients are solved for and converted into the coefficient of lift and drag with the ratio calculated.
- Next, for the subsonic case, the equations for Stratford's flow separation criteria are solved for:
    - One thing thin airfoil theory does not take into account 1) The minimum pressure coefficient that forms at the leading-edge radius of the wing for the upper surface dictated by the leading-edge suction-pressure coefficient formula. 2) The stagnation pressure where the angle of the upper surface is high compared to the angle of the freestream angle.
    - Next, the stagnation effects are added to the thin airfoil theory coefficient of pressure that is used to obtain the minimum pressure coefficient along the upper surface. This is used in the equations for Stratford's separation criteria.
    - The canonical pressure coefficient is then solved for that feeds into Stratford's separation equations.
- Next, the lift to drag ratio is compared for each combination of geometric variables of the airfoil. The highest lift to drag ratio is the output airfoil.